

L. M. C. Gato

Assistant Lecturer.

A. F. de O. Falcão

Professor.

Department of Mechanical Engineering,
Instituto Superior Técnico,
1096 Lisboa Codex, Portugal

On the Theory of the Wells Turbine

A theoretical investigation is presented concerning the aerodynamic performance of the Wells turbine, a self-rectifying, axial-flow turbine suitable for energy extraction from a reciprocating air flow. A two-dimensional analysis is developed, and expressions, based on potential flow, are derived for the blade shape maximizing the turbine efficiency. Three-dimensional effects and profile losses are then accounted for by means of an actuator disk theory, which shows that large radial distortions of axial velocity profile can occur, depending on blade shape, with important implications on the extent of the stall-free conditions.

1 Introduction

A problem, encountered in some devices for sea-wave energy conversion, consists in transforming a reciprocating flow of air into unidirectional rotational motion, which can be used to drive an electrical generator. This requirement is met, without the need of any rectifying valves, by the turbine invented some years ago by Dr. A. A. Wells. It is an axial-flow turbine, constituted essentially by a rotor with untwisted aerofoil blades of symmetrical cross section, set radially at a 90-deg angle of stagger. Versions with and without guide vanes are known to have been tested. The main feature of the turbine is its capability of producing a time-averaged positive power from a cyclically reversing air flow with a fairly high efficiency and a large blade-to-air velocity ratio.

The performance of the Wells turbine has been object of theoretical investigations using the two-dimensional cascade flow model [1-3]. Experimental data have been published only for turbines with rotor blades of constant chord. Grant et al. [2, 4] used a turbine of 0.4-m rotor dia, with guide vanes on both sides of the rotor, whereas Raghunathan et al. [3, 5, 6] tested a turbine of 0.2-m dia without guide vanes.

The main purpose of this paper is to develop a more comprehensive and exact analysis, including three-dimensional effects, and to find out how the blade shape affects the turbine performance. The turbine is supposed to have long cylindrical walls and no guide vanes, and the flow is considered incompressible. A two-dimensional analysis is first developed, in section 2, and expressions, based on potential flow, are derived for the blade shape, allowing maximum efficiency for given values of the hub-to-tip diameter ratio, flow rate coefficient, and exit-loss coefficient. Three-dimensional effects and profile losses are considered, in section 3, by means of an actuator disk flow model, which is used to obtain numerical results showing the effect of blade shape upon turbine performance.

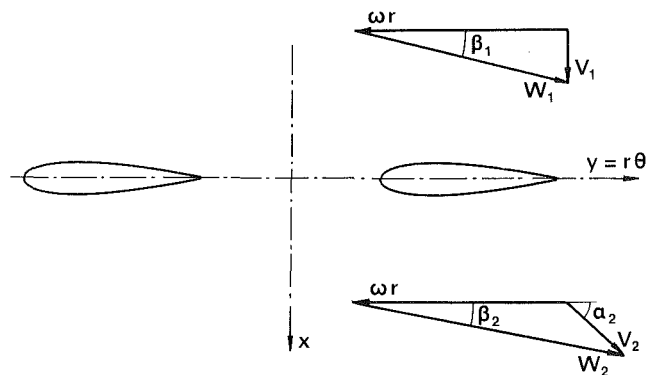


Fig. 1 Two-dimensional cascade geometry

2 Two-Dimensional Analysis

2.1 Basic Assumptions. We introduce a system of cylindrical coordinates (r, θ, x) and consider the annular flow through the turbine to be incompressible and bounded by two coaxial walls of revolution. For the sake of simplicity of the analysis, the duct enclosing the rotor blades is assumed to have cylindrical walls, at $r = R_i$, $r = R$ ($R_i < R$), long enough so that we may ignore the disturbing end-effects upon the flow pattern about the blades. The rotor has angular speed, ω , and comprises a set of N untwisted blades, of chord, $c(r)$, and symmetrical profile, located radially at the plane $x = 0$. In the present analysis, the effect of the annular wall boundary layers is neglected, and the incoming flow is supposed uniform and swirl-free, with positive axial velocity, U .

As a first approximation, we ignore the flow interference between different radii, which means that the radial velocities are neglected. In this case, the flow at each cylindrical streamsurface may be adequately represented by the two-dimensional flow about a rectilinear cascade of blades in the (x, y) plane, where we take $y = r\theta$.

We consider, in the (x, y) plane, a cascade of symmetrical aerofoils set at 90-deg angle of stagger along the y -axis (tandem cascade), with chord, c , and pitch, $t = 2\pi r/N$, and moving with velocity, ωr , along the negative y -direction (Fig.

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1). The incoming flow velocity, $\mathbf{W}_1$, relative to the blades, has components $W_{1x} = U$ and $W_{1y} = \omega r$. Far downstream, it is $W_{2x} = U$ and $W_{2y} = \omega r + V_{2y}$. We define the lift and drag directions as perpendicular and parallel, respectively, to the relative mean velocity vector, $\mathbf{W}_m = (\mathbf{W}_1 + \mathbf{W}_2)/2$, and introduce the lift and drag coefficients, c_L and c_D , referred to the relative mean flow dynamic pressure. It can easily be found that the work, $E(r)$, done by the fluid on the blades, per unit mass of fluid, is given by

$$E = \frac{1}{2} c_L \frac{c}{t} \omega r U \frac{\sin(\beta_m - \gamma)}{\sin^2 \beta_m \cos \gamma} \quad (1)$$

where $\gamma = \arctan(c_D/c_L)$, and β_m is the angle of $\mathbf{W}_m$ with the y -direction. Applying the momentum theorem, we obtain

$$\tan \beta_m = \frac{U}{\omega r + \frac{E}{2\omega r}} \quad (2)$$

which, together with equation (1), allows β_m and E to be determined if c_L and c_D are known as functions of β_m (or β_1). The pressure drop through the cascade is then given by

$$p_1 - p_2 = \frac{1}{2} c_L \frac{c}{t} \rho U^2 \frac{\cos(\beta_m - \gamma)}{\sin^2 \beta_m \cos \gamma} \quad (3)$$

In order to define the turbine efficiency, η , we assume that the swirl kinetic energy at exit is lost, as well as a fraction $KU^2/2$ ($K \leq 1$) of the axial-velocity kinetic energy. It follows that $\eta = P/T$, where

$$\left. \begin{aligned} P &= 2\pi\rho U \int_{R_i}^R E(r) r dr \quad \text{and} \\ T &= 2\pi U \int_{R_i}^R \left(p_1 - p_2 + K\rho \frac{U^2}{2} \right) r dr \end{aligned} \right\} \quad (4)$$

are the power output of, and the power available to, the turbine, respectively.

In unstalled conditions, β_m is a small angle, of the order of magnitude of γ , which means that the work, E , is expected to depend strongly on the drag coefficient, more markedly than in conventional cascade flows. (In particular, it is $E < 0$ if $\beta_m < \gamma$.) Experimental data on tandem-cascade flow are relatively scarce. The only published results [7] were obtained for a Reynolds number of about 10^5 , which is one to two

orders of magnitude smaller than what is expected to occur in full-scale Wells turbines for sea wave-energy conversion. Aerofoil drag coefficients, and to a lesser extent lift coefficients, are known to change substantially for large Reynolds number variations over these ranges [8], which means that those data are probably unreliable to predict the performance of full-scale turbines. An alternative approach, which will be followed here, consists in using isolated aerofoil experimental data, modified by interference factors derived from cascade potential flow theory. Although this method still does not allow an accurate evaluation of the drag coefficient (which could be affected by interference between aerofoils in an unpredictable manner), it constitutes a fairly reliable way of determining the lift coefficient for a large variety of aerofoil profiles and a wide range of Reynolds numbers, provided the chord-to-pitch ratio is not too close to unity.

We introduce the cascade interference factor, $k = c_L/c_{Lo}$, where c_{Lo} is the isolated aerofoil lift coefficient at the angle of incidence, β_m , and assume that, for the geometries and Reynolds numbers of interest, k is not substantially affected by viscous and thickness effects. Consequently, we will take, for k , the corresponding value calculated for a cascade of flat plates in potential flow. An exact solution for the plane potential flow about an infinite row of flat plates with arbitrary angle of stagger is well known [9–11], and the theoretical value of the interference factor is found to be independent of the angle of incidence, β_m . For the special case of 90-deg angle of stagger, it is simply

$$k = \frac{2t}{\pi c} \tan \frac{\pi c}{2t} \quad (5)$$

2.2 Potential Flow. Before dealing with the case of aerofoils in real flow, we derive some theoretical results for the potential flow about blades of negligible thickness (flat plates). In this case, it is $\gamma = 0$, and $c_{Lo} = 2\pi \sin \beta_m$. The angle α_2 , of the downstream velocity $\mathbf{V}_2$, is given by

$$\tan \alpha_2 = \frac{U}{V_{2y}} = \frac{1}{2} \cotan \frac{\pi c}{2t} \quad (6)$$

which shows α_2 to depend only on the cascade solidity. Introducing the dimensionless quantities $[r^*, R^*, c^*] = [r, R, c]/R$, $U^* = U/(\omega R)$, $[P^*, T^*] = [P, T]/(\rho\omega^3 R^5)$, we obtain, for the efficiency, $\eta = P^*/T^*$, where $T^* = P^* + L_x^* + L_y^*$

$$P^* = 4\pi U^{*2} \int_{R^*}^1 r^{*2} \tan \frac{\pi c^*}{4r^*} dr^* \quad (7)$$

Nomenclature

c = blade chord	(r, θ, x) = cylindrical coordinate system	ω = rotor angular speed
c_D, c_L = drag, lift coefficient	R_i, R = inner, outer radius	Subscripts
c_{Lo} = isolated aerofoil lift coefficient	S = solidity, total blade area/annular area	
E = blade work per unit mass of fluid	t = cascade pitch	av = time-averaged value in oscillating flow
k = cascade interference factor	T = power available to the turbine	o = conditions of maximum flow rate in oscillating flow
K = fraction of axial kinetic energy lost at exit	U = inlet flow velocity	op = optimum value for maximum efficiency
L_x, L_y = exit energy losses due to axial, tangential velocity	v = velocity perturbation to radial equilibrium	1, 2 = upstream, downstream of blade row
N = number of rotor blades	V, W = absolute, relative flow velocity	$+\infty, -\infty$ = radial equilibrium conditions, at $x = \pm\infty$
p = pressure	(x, y) = cartesian coordinate system (Fig. 1)	Superscripts
$\Delta p, \Delta p_o$ = drop in static, stagnation pressure across the disk	α, β = angle of absolute, relative velocity	
P = turbine power output	γ = $\arctan(c_D/c_L)$	$*$ = dimensionless value
Q = volume flow rate	η = turbine efficiency	$-$ = time-averaged value in oscillating flow
	ρ = density	

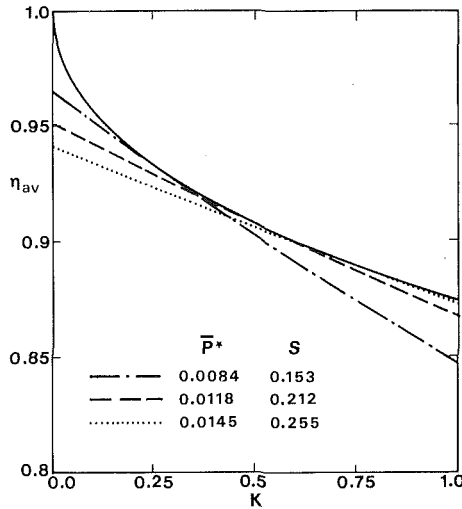


Fig. 2 Results from two-dimensional potential flow theory for blades of optimum shape

$$L_x^* = \frac{\pi}{2} K(1 - R^{*2})U^*, L_y^* = 4\pi U^{*3} \int_{R^*}^1 r^* \tan^2 \frac{Nc^*}{4r^*} dr^* \quad (8)$$

L_x^* and L_y^* being dimensionless coefficients for the exit losses due to axial and swirl velocity components, respectively.

Since the Wells turbine is intended primarily for applications with oscillating flow, it is convenient, at this stage, to introduce time-averaged values. Assuming sinusoidally varying flow rate, we may write $U(t) = U_o \sin \Omega t$. Then, since the average values of $\sin^2 x$ and $\sin^3 |x|$ over a cycle are equal, respectively, to $1/2$ and $4/(3\pi)$, it follows that $\bar{P}^* = P_o^*/2$ and $[\bar{L}_x^*, \bar{L}_y^*] = [L_{xo}^*, L_{yo}^*]4/(3\pi)$, where the superscript bar and the subscript, o , indicate time-averaged and maximum flow rate conditions, respectively. The following relation holds for the corresponding efficiencies

$$\eta_{av} = \frac{\bar{P}^*}{\bar{T}^*} = \left[1 + \frac{8}{3\pi} \left(\frac{1}{\eta_o} - 1 \right) \right]^{-1} \quad (9)$$

from which it can be found that $\eta_{av} > \eta_o$, a result which should be expected, since η decays with increasing flow rate as a consequence of the losses rising faster than the power produced (see equations (7) and (8)). (Note that η_{av} is not the time-averaged value of η .)

For given values of R^* , U^* , and K , η can be considered as a functional of $\mu(r^*) = Nc^*(r^*)$, and to find the function $\mu(r^*)$ that gives a maximum of η is a problem of variational calculus. We shall deal with this question first by looking for the function $\mu(r^*)$ that yields a minimum of L_y^* (i.e., a maximum of η) for fixed P^* . This is the classical isoperimetric problem (see e.g. [12]) and its solution in this case can easily be found to be

$$Nc^* = 4r^* \arctan \frac{r^* P^*}{\pi U^{*2}(1 - R^{*4})} \quad (10)$$

from which we obtain

$$L_y^* = \frac{P^{*2}}{\pi U^{*2}(1 - R^{*4})} \quad (11)$$

Equation (10) gives the optimum chord distribution for a given power coefficient, P^* . (This will be referred to hereafter as blade shape A.) From equation (11), we obtain the expression of the function $\eta = f(P^*)$ giving the maximum efficiency for each value of P^* . Then, by setting $f'(P^*) = 0$, we find the optimum value of P^*

$$P_{op}^* = \pi U^{*2}(1 - R^{*2}) \left[\frac{K}{2} (1 + R^{*2}) \right]^{1/2} \quad (12)$$

Table 1 Highest efficiency conditions for several blade shapes

	S	$\bar{P}_{op}^*$	$(\eta_{av})_{max}$
Shape A	0.212	0.0118	0.908
$c/r = \text{const}$	0.216	0.0116	0.906
Shape B	0.212	0.0110	0.901
$c = \text{const}$	0.208	0.0108	0.901

which turns out to give $L_x^* = L_y^*$, and finally yields the following expression for the maximum value, η_{max} , of the efficiency η , for fixed R^* , U^* , and K

$$\eta_{max} = \left[1 + U^* \left(\frac{2K}{1 + R^{*2}} \right)^{1/2} \right]^{-1} \quad (13)$$

The time-averaged efficiency, η_{av} , in oscillating flow, is plotted in Fig. 2 versus the loss coefficient K , for $R^* = 0.5$, $U_o^* = 0.134$, and several values of $\bar{P}^* = P_o^*/2$ (or of the solidity S , i.e. total blade area divided by annular area), it being assumed that the radial distribution of the chord satisfies equation (10) for each value of P_o^* (blade shape A). In the same figure, the solid line represents the maximum time-averaged efficiency, $(\eta_{av})_{max}$, as given by equations (13) and (9), and may be considered as the envelope of the curves for each value of P_o^* .

It is to be noted that the power coefficient, P_{op}^* , for maximum efficiency, increases with K (see equation (12)), and also that it is $(\eta_{av})_{max} = 1$ for $K = 0$, but the latter condition implies vanishingly small values of P^* and S .

Equations (10–13) define the blade shape and solidity, and the turbine performance, for fixed R^* , U^* , and K , if maximizing the efficiency is the dominating design condition. It is of interest to know how the efficiency is affected if other blade shapes are chosen instead, which may be more convenient from other points of view, namely manufacturing and blade stresses. It can easily be found that, instead of equation (13), we have

$$\eta_{max} = \left[1 + \frac{3}{2} U^* \sqrt{K} \frac{1 - R^{*2}}{1 - R^{*3}} \right]^{-1}$$

for $c/r = \text{constant}$ (constant chord-to-pitch ratio), and

$$\eta_{max} = \left[1 + 2U^* \left(\frac{K}{2} \frac{\ln R^*}{R^{*2} - 1} \right)^{1/2} \right]^{-1}$$

for constant blade work, E (which will be referred to as shape B and gives c increasing with r). Table 1 compares these cases, and also the turbine with constant blade chord, for $K = 0.5$, $U_o^* = 0.134$, and $R^* = 0.5$. The results show the maximum attainable efficiency, and the optimum power coefficient and solidity, to be only slightly affected by the blade shape. We recall that this conclusion is based on two-dimensional, blade-element inviscid flow model. More marked differences are found if three-dimensional effects and profile losses are included in the analysis, as shown in section 3.

3 Three-Dimensional Analysis

3.1 Actuator Disk Theory. The assumption of zero radial velocity, which is a condition for the validity of the two-dimensional analysis, is exact only in the special case of incompressible irrotational flow and constant blade circulation (or constant blade work). In the general case of blade circulation varying with the radius, the shed vorticity induces radial velocities which make the stream surfaces deviate from the circular cylindrical shape. In order to account for this effect, we use the well known actuator-disk flow model, which assumes the shed vorticity to be axisymmetrically distributed [13]. Moreover, we consider the shed vortex lines to have approximately the shape of semi-infinite true helical

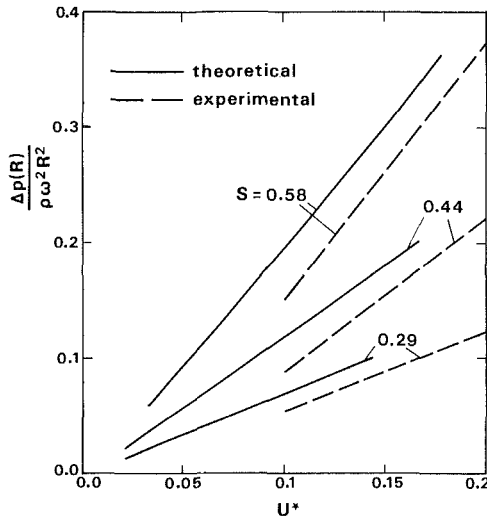


Fig. 3 Theoretical and experimental pressure drop across blade row versus inlet flow velocity

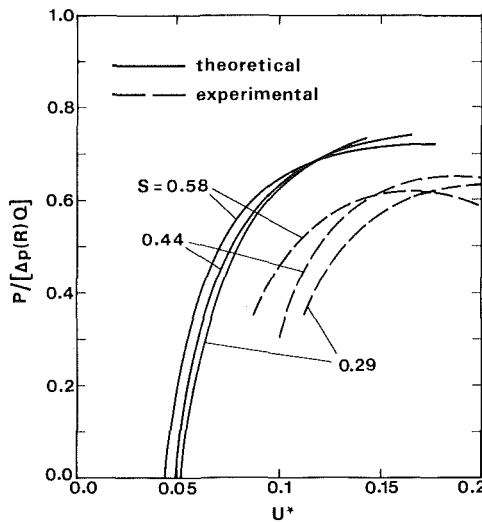


Fig. 4 Theoretical and experimental rotor efficiency versus inlet flow velocity

lines (linear actuator disk theory). This requires the radial velocity, and hence the shed vorticity, to be everywhere small, which in turn implies that the blade circulation should not be allowed to vary strongly with the radius. (We will however apply our analysis to cases when this condition is not satisfied, the consequence being that the results shall not then be expected to be fully accurate.) The actuator disk equations yield the (radially varying) axial velocity at the disk, which will then allow the flow deflection and the pressure drop to be known from two-dimensional cascade data.

We assume inviscid, incompressible, axisymmetric flow and write, for the velocity $\mathbf{V}$ (with radial, tangential and axial components V_r, V_θ, V_x),

$$\mathbf{V} = \mathbf{V}_{\pm\infty} + \mathbf{v} \text{ for } x \gtrless 0, \quad (14)$$

$\mathbf{V}_{\pm\infty}$ being the velocity far away from the blade row, at $x = \pm\infty$, where the flow is in radial equilibrium and so $V_{-\infty x} = U = \text{constant}$, $V_{-\infty\theta} = V_{-\infty r} = V_{+\infty r} = 0$. The velocity, $\mathbf{v}$, represents the perturbation from radial equilibrium flow, and, under the assumptions above, is known to be an irrotational vector field [13].

The continuity equation, together with the tangential component of $\nabla \times \mathbf{v} = 0$, gives the following differential equation for $v_x(r, x)$

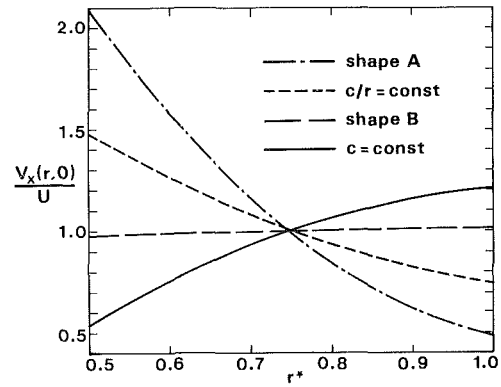


Fig. 5 Axial velocity profile at disk for several blade shapes

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_x}{\partial r} \right) + \frac{\partial^2 v_x}{\partial x^2} = 0$$

a solution of which is

$$v_x(r, x) = \pm \sum_{n=1}^{\infty} A_n e^{\mp \lambda_n x} F_n(r) \text{ for } x \gtrless 0 \quad (15)$$

where $F_n(r) = J_0(\lambda_n r) + b_n Y_0(\lambda_n r)$, J_0 and Y_0 being Bessel functions. The eigenvalues λ_n , b_n ($n=1, 2, \dots$) can be found from the boundary condition $v_r = 0$, or equivalently $F'_n(r) = 0$, for $r = R_i$ and R . The condition of continuous axial velocity across the disk yields

$$v_x(r, -0) = -v_x(r, +0) = \frac{1}{2} [V_{+\infty x}(r) - U] \quad (16a)$$

$$= - \sum_{n=1}^{\infty} A_n F_n(r). \quad (16b)$$

In order to solve the problem, we consider the radial component of Euler equation

$$V_r \frac{\partial V_r}{\partial r} + V_x \frac{\partial V_r}{\partial x} - \frac{V_\theta^2}{r} = - \frac{1}{\rho} \frac{\partial p}{\partial r}$$

which can be written, at the exit from the disk, $x = +0$, as

$$\frac{d}{dr} v_x(r, +0) = \frac{1}{2 V_x(r, 0)} \frac{V_\theta(r, +0)}{r} \frac{d}{dr} [r V_\theta(r, +0)] - \frac{1}{\rho} \frac{d}{dr} \Delta p_o \quad (17)$$

where Δp_o is the drop in stagnation pressure across the disk. A solution is then worked out iteratively as follows. We start by assuming a radial distribution for the axial velocity at the disk, $V_x(r, 0) = U + v_x(r, -0)$. From two-dimensional cascade data, we obtain Δp_o , and also the swirl velocity downstream of the cascade, which we take as the value of $V_\theta(r, +0)$. Then equation (17) allows $dv_x(r, +0)/dr$ to be known as a function of r . The coefficients A_n can now be computed by taking into account that

$$\frac{d}{dr} v_x(r, +0) = - \sum_{n=1}^{\infty} A_n \lambda_n [J_1(\lambda_n r) + b_n Y_1(\lambda_n r)]$$

and using the orthogonality properties of the Bessel functions. So, from equation (16b), we obtain the new value of $V_x(r, 0)$ which completes the iterative cycle.

3.2 Numerical Results. Figures 3 and 4 show a comparison between results from our actuator disk analysis and experimental data obtained by Raghunathan and Tan [6] for a turbine with $R = 0.1$ m, $R^* = 0.62$ and blade chord in-

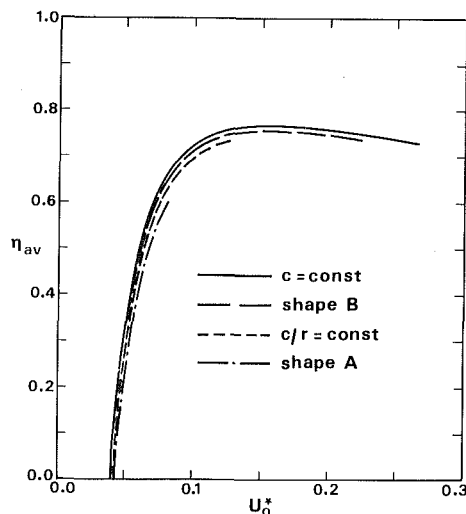


Fig. 6 Time-averaged turbine efficiency in oscillating flow versus inlet velocity amplitude, for several blade shapes

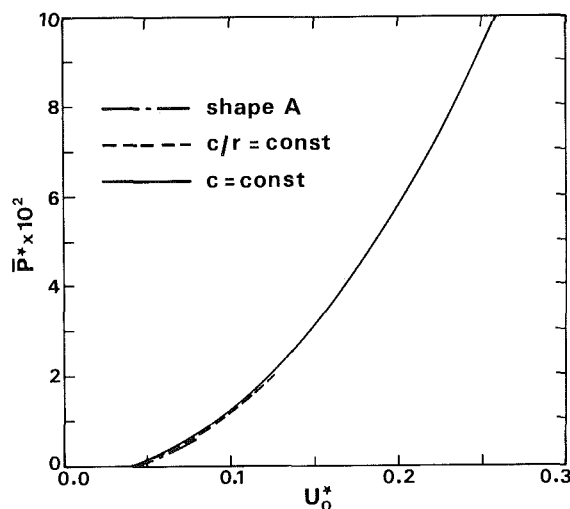


Fig. 7 Time-averaged power output in oscillating flow versus inlet velocity amplitude for several blade shapes

dependent of radius. The experiments were performed with blade profile NACA 0021 at Reynolds number of about 2.3×10^5 (based on relative velocity at blade tip). The values of c_{Lo} and c_D for the calculations were taken from [8], for NACA 0018 (since no available data were found for NACA 0021 at such low Reynolds numbers). Figure 3 represents the variation of the static pressure drop across the blade row (or disk), $\Delta p(R)$, measured or calculated at the outer wall, versus inlet flow velocity. Figure 4 shows a plot of the efficiency, defined here as $P/[\Delta p(R)Q]$ (i.e., ignoring exit losses). In both figures, each theoretical curve ends at the value of U_o^* for which the maximum value of $\beta_m(r)$ equals about 11 deg; this value, for the corresponding Reynolds number and isolated aerofoil, is the highest angle of incidence for stall-free conditions (see [8]). The discrepancies between theoretical and experimental results are possibly due, in part, to joint effects of the following factors.

(a) Annular-wall boundary layers, blade-tip clearance and mechanical losses (all of which are believed to affect significantly the performance of a small turbine) are not accounted for by the theory.

(b) The theoretical calculations were based on c_D and c_{Lo} data for NACA 0018 instead of 0021. (For $Re \sim 2 \times 10^5$, c_{Lo} is known to decrease with increasing relative thickness, see [8].) In addition, we mention again that no attempt was made in

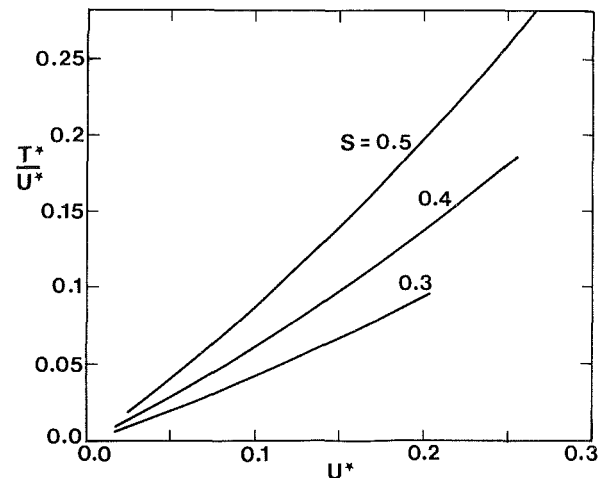


Fig. 8 Dimensionless plot of overall pressure drop across the turbine versus flow rate, for constant-chord blades

the analysis to correct the drag coefficient for cascade interference effects.

The results plotted in Figs. 5–8 were based on aerofoil data from [8] for NACA 0018 and $Re \sim 10^7$ (considered representative of a full-scale industrial turbine), and computed for $R^* = 0.5$, $K = 0.5$. Figures 5–7 illustrate the effect of blade shape upon turbine performance, for constant solidity, $S = 0.5$. Figure 5 shows the radial distribution of the axial velocity component at the disk. Blades of shape A (defined as in section 2.2) and, less strongly, of $c/r = \text{const}$, induce a distortion in axial velocity profile that tends to aggravate the appearance of blade stall at the hub (in comparison with what should be expected from two-dimensional analysis), whereas constant-chord blades perform favorably in this respect. This effect is clearly illustrated in Fig. 6, where the efficiency curves for different blades shapes terminate at different values of U_o^* (which are taken as the limit of stall-free conditions and are those for which the maximum value of $\beta_m(r)$ equals 16 deg).

Figure 6 shows a plot of the time-averaged efficiency, η_{av} , for sinusoidally varying flow rate, versus (dimensionless) amplitude of the inlet flow velocity. Here, we define, as before, $\eta_{av} = \bar{P}/\bar{T}$, where now it is

$$T = 2\pi \int_{R_i}^R V_x(r, 0) \left[\Delta p(r) + \frac{K\rho}{2} V_x^2(r, 0) \right] r dr$$

A similar plot is given in Fig. 7 for the time-averaged power output. It is interesting to notice that the positions of the various blade shapes, on what concerns efficiency, as shown in Fig. 6, are inverted in comparison with the results of Table 1, where three-dimensional effects and profile losses are not accounted for.

The Wells turbine has often been referred to as satisfying a relationship of proportionality between flow rate and overall pressure difference (or gross available energy per unit mass of fluid). Figure 8 shows this to be only approximately true.

4 Conclusions

The results of the theoretical analysis show that fairly high values of the mean efficiency, of the order of 70–80 percent, are expected to be attainable in reciprocating flow with properly designed Wells turbines.

Three-dimensional flow effects, which were ignored in previous analyses, are found to be large, depending strongly on blade shape, and to be especially important in connection with the extent of the stall-free conditions. Except for blades

which do not differ much from shape B (constant-circulation blades), the present actuator-disk linear analysis is believed to be unable to give an accurate representation of the flow field. This will require a nonlinear, three-dimensional theory.

Finally, we mention that the presently available experimental data concerning tandem cascades of blades are not sufficient as a basis for an accurate prediction of the performance, and a reliable design, of full-scale Wells turbines. Steps towards improving this situation are desirable.

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